ESCOLA
SUPERIOR
DE MEDIA
ARTES
E DESIGN
POLITÉCNICO
DO PORTO

ALP

FICHA DE EXERCÍCIOS

ATIVIDADE LETIVA

Algorithmics and Data Structures

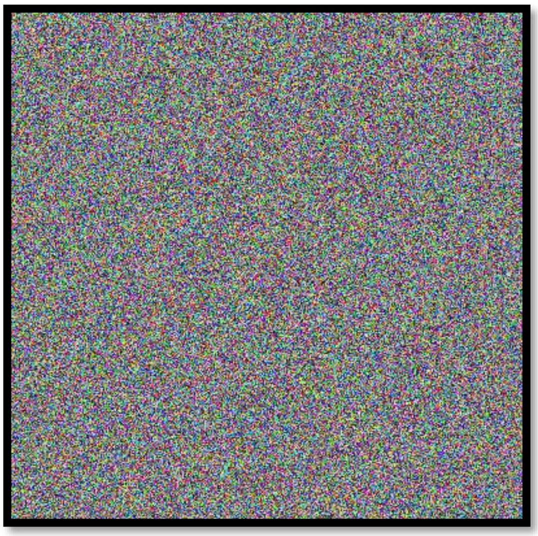
UNIDADE CURRICULAR

Sheet 06 – Two-dimensional lists applied to images – PIL library

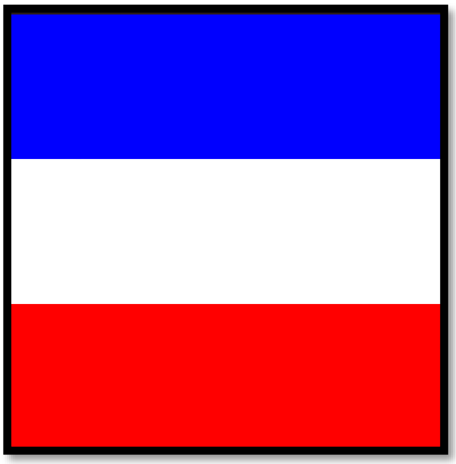
1. Develop an **ImageArt() function** that creates an image of (400, 400) pixels, in RGB mode.

   The function should fill the pixel map of the image with randomly generated RGB (**Red**, **Green**, **Blue**). Each of them between 0 and 255.

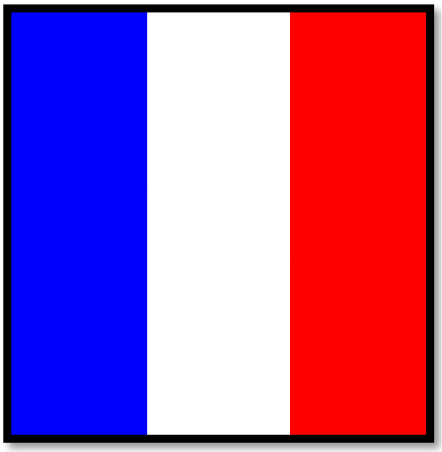
   Show the image and save it, with the name **imageArt.jpg** , to an image subfolder.



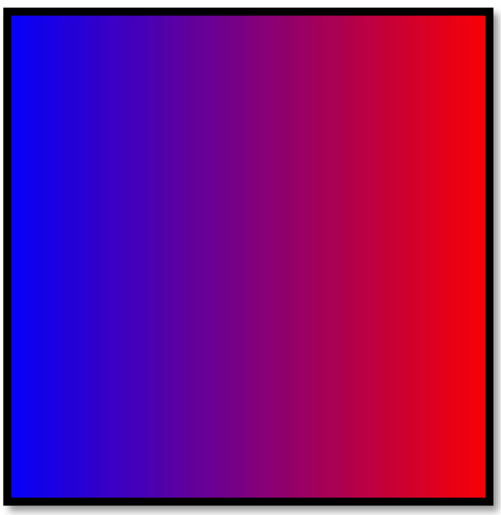
2. Look for the example available in Moodle (PILLOW presentation) that creates an image (240, 240) with 3 stripes each of them 80 pixels high (blue, white, red).



3. Do a similar function, but create 3 vertical stripes, of 80 pixels each. Save the image in the images subfolder, with the name **flag.jpg**



4. Create a function imageGradiente() that creates an image of (250, 250) pixels, in RGB mode. The image should represent a smooth gradient from blue to red (no green):



5. Create an **imageFrame()** function that takes an image as the input argument, and creates a frame around it with 20 pixels in magenta, as illustrated below.

6. Create an **imageGrayScale**() function, which receives an image and creates a new image (from the received one) in GrayScale, but using an improved formula compared to the one shown above: RBG of each pixel = (Red + Green + Blue)/3.

   Thus, the new NTSC GrayScale formula in which each pixel = **0.299 * Red + 0.587 * Green + 0.114 * Blue** better represents the human perception of luminosity and brightness in Gray Scale images.

7. Create a function that receives an image and create a window-shaped frame with 10 pix-els in blue.

8. Create a function than receives an image and create another image with a rotation passed as argument of the function.